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Prof. Jin Xuan and Prof. Jinfeng Liu
Editors-in-Chief
Digital Chemical Engineering
Elsevier

Dear Prof. Xuan and Prof. Liu,

Please find enclosed the revised manuscript, "An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity" (DCHE-D-26-00020R1), for consideration in *Digital Chemical Engineering*.

We are grateful for the reviewer's careful and constructive assessment. We have addressed all remaining comments and revised the manuscript to explain the method more clearly, present its physical guarantees accurately, and describe its predictive performance with appropriate balance. A detailed point-by-point response accompanies the manuscript.

We believe the article will interest readers of *Digital Chemical Engineering* because it presents an interpretable and practical surrogate for activated-sludge systems. The approach produces physically valid deployed predictions, performs well with limited training data, and is evaluated transparently against established machine-learning models. It therefore offers a useful balance of interpretability, reliability, and computational efficiency for wastewater-process analysis and decision-making.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The authors declare no conflicts of interest. Thank you for considering this revised work for publication in *Digital Chemical Engineering*.

Sincerely,

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